

# Quant, Systematic & Algo Trading

Part of The Complete Indian Market Trader — Zero to Professional (India, 2026)

Pro-grade, India-first: NSE/BSE, F&O, worked ₹ examples, current 2026 rules

## Contents

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1. **From Discretionary to Systematic Trading**
2. **Designing a Trading Strategy**
3. **Backtesting Properly (Bias, Costs, Walk-Forward)**
4. **Performance Metrics That Matter**
5. **Systematic Strategy Families**
6. **Position Sizing & Portfolio of Strategies**
7. **Python for Algo Trading**
8. **Execution, Broker APIs & the SEBI Retail-Algo Framework (2026)**
9. **Deploying & Running a Live Algo**

# From Discretionary to Systematic Trading

## Why this matters — the pro vs retail gap this closes

You already know candlesticks, Greeks, and 200 option structures. That knowledge is necessary but not sufficient. The gap that actually separates a profitable pro from the 90% of retail F&O traders who lose money (SEBI's own study covering FY22–24 pegged net losers at roughly 91–93% of individual F&O traders, average loss around Rs 1.1–1.2 lakh) is not *more* setups. It is **repeatability**. A retail trader who "reads the chart" makes a different decision on the same Bank Nifty setup depending on whether he is up or down for the day, whether he slept well, and whether the last trade won. A systematic trader takes the same input and produces the same output every time. That consistency is what lets you measure an edge, size it, and compound it.

This chapter reframes everything you already know into a process you can test, trust, and scale. It does not throw away discretion — it disciplines it.

## The essentials — edge, randomness, and the three modes

**Edge vs randomness.** An "edge" is a positive statistical expectancy after all costs: over many trades,  $(Win\% \times Avg\ Win) - (Loss\% \times Avg\ Loss) - Costs > 0$ . Any single trade tells you nothing — a coin can land heads five times. You need a *sample*. As a rough rule, you cannot distinguish a real edge from noise on fewer than ~100 independent trades, and ideally 200–300. This is why one great week means nothing and why screenshots of single trades are the marketing tool of people who lose over the year.

### The three modes of trading:

| Mode             | Decision source                                    | Repeatability | India-specific reality (2026)                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------|----------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Discretionary    | Human judgement per trade                          | Low           | Legal, flexible, but unmeasurable and emotion-prone                                                                                                                                                                                                                                                                                                                                                                     |
| Systematic       | Fixed written rules, executed by hand or semi-auto | High          | You decide; rules are backtestable                                                                                                                                                                                                                                                                                                                                                                                      |
| Algo (automated) | Code places orders                                 | Highest       | Under <b>SEBI Retail Algo Framework (mandatory 01-Apr-2026)</b> : every algo order needs an <b>exchange Algo-ID</b> , <b>open APIs are banned</b> , retail algos run <b>only through a registered broker's authenticated API</b> , and orders above <b>10/sec</b> need exchange registration. Third-party strategy vendors must tie up with a registered broker. <i>Verify current text on NSE/SEBI — rules change.</i> |

The key insight: **systematic and algo are the same logic; only the executor differs**. Build the system first as written rules a human can follow, prove the edge, and only then consider automating it inside the broker-API rules above. Automating an unproven system just loses money faster.

**Why rules beat gut over time.** Gut feel is pattern recognition trained on a biased, tiny, emotionally-weighted sample. It is genuinely good at some things (reading order-flow context, sensing an event day) and terrible at others (position sizing, cutting losers, avoiding revenge trades). Rules are unemotional, auditable, and improvable. When a rule loses, you can inspect *why*; when your gut loses, you just feel bad.

## Worked example — same setup, two traders

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Setup: Bank Nifty (lot size 35 as of 2026 — *verify with NSE, lot sizes revise*) opens and forms a 15-minute opening range. A common rule: **buy the future on a breakout above the first 15-min high, stop at the range low, target 2× risk.**

Say Bank Nifty's first 15-min range on a given day is 52,000–52,180 (180 points wide). Breakout entry 52,180, stop 52,000 (risk 180 points), target 52,540 (+360).

- **Per-point value:** Rs 35/point (1 lot). Risk =  $180 \times 35 = \text{Rs } 6,300 \text{ per lot}$ . Reward if target hit =  $360 \times 35 = \text{Rs } 12,600$ .
- **Costs (round trip, 2026, one lot future ~Rs 18.2 lakh notional):** brokerage ~Rs 40 (Rs 20 × 2 legs, flat-fee broker), STT on futures ~0.05% on sell side ≈ Rs 910 on ~Rs 18.2L sell value, exchange txn + SEBI + stamp + 18% GST on (brokerage+txn) stack to roughly Rs 150–250 more. Call it **~Rs 1,200–1,400 all-in per lot round trip**. *STT rates effective 01-Apr-2026 per Budget 2026 — verify on your broker's contract note.*

The **systematic trader** takes every qualifying breakout, logs it, and after 150 trades knows his real win rate (say 38%) and payoff (avg win 1.9R, avg loss 1.0R). Expectancy per trade =  $0.38 \times 1.9 - 0.62 \times 1.0 = 0.722 - 0.62 = +0.10R$  before costs — thin, and costs of ~Rs 1,300 on a 180-point (Rs 6,300) risk eat ~0.21R. **Net expectancy is negative.** The system fails the honesty test, and he *knows it* — so he widens the target, trades bigger ranges, or drops the strategy.

The **discretionary trader** takes the breakout when he "feels" it, skips it when he's scared, doubles up after a loss, and never logs anything. He cannot compute any of the above. He will discover the negative edge only when his account is gone.

The lesson isn't that breakouts are bad — it's that only the systematic trader could *see the truth* and act on it.

## How pros do it / common mistakes

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**How pros do it:** - **Write the rule before the trade, not after.** If it isn't written, it isn't a system. - **Trade a sample, then judge.** No conclusions before ~100 trades. - **Cost-first thinking.** Pros subtract 2026 STT + charges *before* deciding a setup is worth it. Many "edges" are real gross and negative net. - **One change at a time.** Change a stop OR a filter, never both, so you can attribute results. - **Keep a trade journal that a stranger could audit** — entry, exit, size, R, reason, screenshot.

**Classic retail errors / red flags:** - Believing a 5-trade winning streak is skill. - Optimising rules on a chart you've already seen (this is curve-fitting — Chapter 3). - Position sizing by "how confident I feel" instead of fixed risk. - Chasing automation before proving the edge — and worse, using a banned open API or an unregistered third-party bot post 01-Apr-2026, which puts your broker relationship and your capital at legal risk. - No log. If you can't compute your expectancy, you don't have a system; you have a hobby that costs money.

**Realistic expectations.** A good, simple, honestly-costed retail system in Indian markets might deliver a Sharpe of 0.8–1.3 and 15–25% annual returns with 15–20% drawdowns — *if* you survive the drawdowns without abandoning it. Anyone promising 5% a month consistently is selling you something. The systematic path is slower and more boring than the YouTube fantasy; that boredom is the point.

## Checklist / drill

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**Systematic-readiness checklist — before you trade any strategy:** - ☐ The full rule is written in one paragraph a stranger could execute. - ☐ Entry, exit, stop, target, and position size are all objective (no "looks strong"). - ☐ I have estimated 2026 all-in costs per round trip for this instrument. - ☐ I have a blank journal ready to log every trade (win or loss). - ☐ I've committed to a minimum sample ( $\geq 100$  trades) before judging. - ☐ I know the max losing streak I'll tolerate before pausing.

**Drill (this week):** Take your favourite discretionary setup. Force it into written rules. Paper-trade or backtest 30 instances and log each in R-multiples. Compute win%, avg win, avg loss, and expectancy *net of 2026 costs*. You will likely be surprised — and that surprise is your first real step from gambler to trader.

# Designing a Trading Strategy

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## Why this matters — the pro vs retail gap this closes

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Retail traders collect *setups*; pros build *systems*. A setup is "buy when RSI is oversold." A system specifies the universe, the exact entry and exit, the filters, the timeframe, the position size, and the costs — every decision pre-made so that execution is mechanical. The difference matters because a vague setup cannot be tested, cannot be sized, and cannot be improved. This chapter turns a trading *idea* (a hypothesis) into a complete, objective, testable **strategy specification**. Getting this right also inoculates you against the single biggest killer of retail quant efforts — curve-fitting — which starts not in the backtest but here, at design time.

## The essentials — from hypothesis to rules

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A strategy is a falsifiable hypothesis plus six mechanical components. Start with the hypothesis in plain English, because it forces you to have an *economic reason* the edge exists.

**1. Hypothesis (the "why").** Example: "*Nifty tends to continue an intraday trend after a strong, wide-range first hour, because institutional order flow persists through the session.*" If you can't state why an edge should exist, you're just data-mining noise.

**2. Universe (what you trade).** Be specific: Nifty 50 index futures? Bank Nifty weekly options? The Nifty 100 cash stocks with average daily turnover > Rs 50 crore? A liquid universe controls slippage. Illiquid mid-caps will destroy a backtested edge in real fills.

**3. Timeframe.** Intraday 5-min? Daily close? This determines your data, your costs (intraday STT is 0.025% on sell for equity; F&O differs), and your tax head — **intraday equity is speculative business income, F&O is non-speculative, delivery is STCG/LTCG (FY2026-27)** — *verify with a CA; rules change.*

**4. Entry rule (objective trigger).** No adjectives. Not "when it looks strong." Instead: "*Enter long at the next candle open when the 20-EMA > 50-EMA AND price closes above the previous day's high.*"

**5. Exit rule(s).** Stop-loss, target, and/or time-based exit. E.g. "*Stop = entry - 1 × ATR(14); target = entry + 2 × ATR; force-exit any open position at 15:15 IST.*" Exits matter more than entries for most systems.

**6. Filters.** Conditions that switch the system off: no trades on RBI policy / Union Budget days; skip if India VIX > 25; only trade 09:30–14:30 to avoid the noisy open and the close. Each filter must earn its place — see curve-fitting below.

**7. Position sizing.** The rule that maps signal to quantity. The professional default is **fixed-fractional risk**: risk a fixed % of equity per trade (commonly 0.5–1%). Position size =  $(\text{Equity} \times \text{Risk}\%) / (\text{Stop distance} \times \text{per-point value})$ , then round *down* to whole lots.

**Objective vs subjective — the acid test:** could two strangers, given your rules and the same chart data, place the identical trade? If yes, it's objective and testable. If no, rewrite it.

## Worked example — a full strategy spec

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**Hypothesis:** Bank Nifty exhibits opening-range momentum; a decisive break of the first 30-min range in the trend direction tends to run.

| Component     | Rule (fully specified)                                                                                                     |
|---------------|----------------------------------------------------------------------------------------------------------------------------|
| Universe      | Bank Nifty index future, current-month, lot size 35 ( <i>verify NSE; lot sizes revise</i> )                                |
| Timeframe     | 5-min candles, intraday only, T+1 settlement irrelevant (no delivery)                                                      |
| Entry         | Long: 5-min close above the 09:15–09:45 range high, only if that range width $\leq 250$ pts. Short: mirror below range low |
| Stop          | Opposite end of the opening range                                                                                          |
| Target        | $1.5 \times$ the range width from entry                                                                                    |
| Filters       | Skip Budget/RBI/monthly-expiry days; skip if India VIX $> 22$ ; no new entries after 13:30; one trade per side per day     |
| Position size | Risk 1% of Rs 5,00,000 = Rs 5,000/trade; size = Rs 5,000 / (range width $\times$ Rs 35)                                    |

**Numeric instance.** Opening range 51,900–52,120 (width 220 pts, within the 250 cap). Long entry on a 5-min close at 52,140. Stop 51,900 (risk 240 pts), target =  $52,140 + 1.5 \times 220 = \mathbf{52,470}$ .

- Risk per lot =  $240 \times 35 = \text{Rs } 8,400$ . Position size =  $\text{Rs } 5,000 / \text{Rs } 8,400 = 0.59$  lots  $\rightarrow$  **round down to 0 lots**. The trade is *too big for 1% risk* on a Rs 5L account. This is the design telling you the truth: either accept  $\sim 1.7\%$  risk to trade 1 lot, or you're undercapitalised for Bank Nifty and should be on Nifty (smaller per-point rupee value) or index options with defined risk.
- If you accept 1 lot: risk Rs 8,400, reward if target hit =  $(52,470 - 52,140) \times 35 = 330 \times 35 = \mathbf{\text{Rs } 11,550}$ , before  $\sim \text{Rs } 1,300$  all-in 2026 costs (STT on futures  $\sim 0.05\%$  sell + brokerage + 18% GST stack). Net reward  $\sim \text{Rs } 10,250$  vs net risk  $\sim \text{Rs } 9,700 \rightarrow$  real payoff  $\sim 1.06R$ , not the  $1.5R$  the target implied. **Costs shrank your edge — that's why they go in the spec, not the appendix.**

## How pros do it / common mistakes

**How pros do it: - Economic rationale first.** Every rule traces back to *why* the edge should exist (order-flow persistence, mean-reversion after overreaction, risk-premium harvesting). Rules with no story are prime suspects for overfitting. - **Fewer parameters, wider zones.** A robust system has 2–4 parameters, and it should work across a *range* of each (a target of  $1.4\times$  or  $1.6\times$  shouldn't flip it from great to terrible). Fragility is the fingerprint of curve-fitting. - **Design the exit and the sizing before the entry.** Amateurs obsess over entries; survival lives in exits and sizing. - **Specify costs and taxes at design time.** Bake in 2026 STT/charges; know your tax head.

**Common mistakes / red flags: - Curve-fitting from the start:** adding a filter "because 2019 would've been better with it," using oddly specific numbers ( $\text{RSI} < 27.5$ , EMA of 43), or a rule that only makes sense for one historical event. If a parameter has a suspiciously precise value, you fit it to the past. - **Subjective words** ("strong," "clearly," "confirmation") that can't be coded. - **Too many filters** — each one you add is another degree of freedom to overfit and fewer trades to test on. - **Sizing by conviction** instead of fixed risk. - **Ignoring liquidity** — a beautiful spec on illiquid stocks dies on slippage.

## Checklist / drill

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**Strategy-spec checklist (all must be YES before backtesting):** - ☐ One-line economic hypothesis for *why* the edge exists. - ☐ Universe named with a liquidity floor. - ☐ Timeframe fixed; tax head and cost model noted (2026 STT/charges). - ☐ Entry, stop, target, time-exit all objective — two strangers get the same trade. - ☐  $\leq 4$  parameters, each with no suspiciously precise value. - ☐ Position sizing = fixed-fractional risk, rounded down to whole lots. - ☐ Every filter has a written justification.

**Drill:** Take one idea and fill the seven-row table above with zero adjectives. Then delete your two "nice-to-have" filters and check the idea still makes economic sense. If removing a filter destroys the logic, the logic was the filter — i.e., you were fitting. Rebuild until the core survives on 3 parameters or fewer.



# Backtesting Properly (Bias, Costs, Walk-Forward)

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## Why this matters — the pro vs retail gap this closes

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A backtest is the closest thing trading has to a laboratory — and the easiest place to lie to yourself. Retail traders run a strategy over past Nifty data, see a smooth rising equity curve, and go live with real money; then the "edge" evaporates. The equity curve wasn't a discovery, it was a *reflection of the very data used to build it*, distorted by hidden biases and missing costs. Pros treat a backtest as an adversarial exercise: they assume it's flattering them and hunt for the reasons why. This chapter covers the four ways a backtest lies — dirty data, look-ahead bias, survivorship bias, and unrealistic costs — and the discipline that catches them: in-sample/out-of-sample splits and walk-forward testing.

## The essentials — the four lies and the two disciplines

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- 1. Data quality.** Garbage in, confident garbage out. Indian-market gotchas: unadjusted prices around **stock splits, bonuses, and dividends** (a 5:1 split looks like an 80% crash if unadjusted), **index reconstitution** (Nifty 50 changes members), holidays and special sessions (Muhurat trading), and bad ticks. Always use split/bonus-**adjusted** series for signals, but remember real trading uses *unadjusted* traded prices for cost/STT calculation.
- 2. Look-ahead bias.** Using information the moment couldn't have had. The classic sins: - Signalling on a candle's **close** but assuming you filled at that same close (you can only act on the *next* bar). - Using the day's high/low to decide an entry that happens *before* the high/low formed. - Using a fundamental number (quarterly result) on its period-end date rather than its *announcement* date. - Survivorship-adjusted or restated data leaking future knowledge. Rule: at decision time  $t$ , the system may only use data available *at or before*  $t$ , and fills happen at  $t+1$  open (or worse).
- 3. Survivorship bias.** Backtesting today's Nifty 50 over ten years tests only the *winners* that survived into the index — the Yes Banks and DHFLs that collapsed are silently excluded, inflating returns. Use a **point-in-time** universe (the actual index constituents on each historical date), or accept your equity-strategy backtest is optimistic.
- 4. Realistic costs and slippage (2026).** This is where most "edges" die. Model every line of the Indian cost stack, effective **01-Apr-2026** (Budget 2026 — *verify on your broker contract note; rules change*):

| Cost                | Equity intraday                   | Equity delivery | Futures        | Options                                |
|---------------------|-----------------------------------|-----------------|----------------|----------------------------------------|
| STT                 | 0.025% on sell                    | 0.1% buy + sell | ~0.05% on sell | ~0.15% on premium (sell) & on exercise |
| Brokerage           | flat ~Rs 20/order (or 0 delivery) | often Rs 0      | ~Rs 20/order   | ~Rs 20/order                           |
| Exchange txn + SEBI | small % of turnover               | "               | "              | higher % on premium                    |
| GST                 | 18% on (brokerage + txn)          | "               | "              | "                                      |
| Stamp duty          | on buy side                       | on buy side     | on buy side    | on buy side                            |
| <b>Slippage</b>     | model it                          | model it        | model it       | model it — worst here                  |

**Slippage** is the gap between your assumed price and your real fill. For liquid Bank Nifty futures, model 1 tick (Rs 0.05 price step, but ~1–2 index points effective); for options and illiquid stocks, model far more. A backtest that fills at the mid-price is fiction — you cross the spread every time.

**Discipline 1 — In-sample (IS) vs out-of-sample (OOS).** Split your history. Build and tune the strategy on the IS portion (say 2015–2021). Then run it *once*, untouched, on the OOS portion (2022–2026). If OOS performance collapses, you overfit. The cardinal rule: **you get to look at the OOS data once**. Every time you peek and re-tune, OOS becomes IS and its honesty is spent.

**Discipline 2 — Walk-forward.** More robust than a single split. Optimise on a rolling window (e.g. 2 years), trade the next 3 months with those frozen parameters, then roll the window forward and repeat. Stitching the out-of-sample 3-month blocks together gives a realistic picture of how re-optimisation would actually have performed. If the walk-forward equity curve is far worse than the fully-optimised one, your edge is mostly curve-fit.

## Worked example — a backtest that lied, then told the truth

You test the Chapter-2 Bank Nifty opening-range system on 2019–2023 5-min data using a simple repo backtester (the project's Nifty/Bank Nifty backtest script — feed it OHLCV, define entry/stop/target functions, get a trade log and equity curve).

**First run (naive):** 620 trades, 44% win, gross CAGR 41%, smooth curve. Looks fantastic. Then you audit:

- **Look-ahead:** entries were filled at the *close* of the breakout candle. Fixing to the *next* 5-min open costs ~2–3 points average per entry. Win rate drops to ~41%.
- **Costs (2026):** ~Rs 1,300 all-in per round-trip lot (STT futures ~0.05% sell + brokerage + 18% GST). On 620 trades that's ~Rs 8.1 lakh of costs. Average gross win was Rs 11,550 and average gross loss Rs 8,400 (from Ch. 2 numbers) — subtract ~Rs 1,300 from *every* trade and expectancy falls hard.
- **Slippage:** add 2 index points (Rs 70/lot) each side → another ~Rs 140/trade.

**Post-audit:** 41% win, avg win ~Rs 10,100 net, avg loss ~Rs 9,800 net. Expectancy =  $0.41 \times 10,100 - 0.59 \times 9,800 = 4,141 - 5,782 = -Rs\ 1,641$  per trade. **The strategy is a net loser.** The naive 41% CAGR was pure look-ahead-plus-zero-cost fantasy. Only the honest backtest saved you Rs 5L+ of live-money tuition.

**The fix path:** widen the target (fewer, bigger wins to overcome fixed costs), add a volatility filter, and re-validate on walk-forward — *not* by tweaking until the same 2019–2023 curve looks pretty again.

## How pros do it / common mistakes

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**How pros do it:** - Assume the backtest is lying and try to break it before the market does. - Reserve OOS data and touch it *once*; prefer walk-forward. - Model the full 2026 cost stack + slippage on the *actual* instrument. - Prefer robustness (works across parameter ranges and sub-periods) over a peak result. - Sanity-check trade count: 30 trades prove nothing; ~200+ across varied regimes (2020 crash, 2021 bull, 2022 chop) is the minimum for belief.

**Common mistakes / red flags:** - A backtest with **no losing months** — nothing real is that clean; you've overfit or leaked future data. - Re-running until OOS looks good (you just converted OOS to IS). - Filling at close, mid, or ignoring slippage. - Testing on today's index members (survivorship). - Optimising 8 parameters on 3 years of data — more knobs than data. - Reporting gross returns and quietly hoping costs "won't be that bad." In 2026 India, costs are *exactly* that bad.

## Checklist / drill

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**Backtest-integrity checklist:** - ☐ Prices split/bonus-adjusted; point-in-time universe (no survivorship). - ☐ Signals use only data available at decision time; fills at next-bar open. - ☐ Full 2026 cost stack modelled (STT/brokerage/txn/SEBI/GST/stamp) + explicit slippage. - ☐ History split into IS and OOS; OOS run exactly once. - ☐ Walk-forward performed; parameters frozen per window. - ☐  $\geq$  ~200 trades spanning bull, bear, and sideways regimes. - ☐ Result robust to  $\pm 10$ –20% parameter changes.

**Drill:** Take any backtest you already believe in. Add Rs 1,300/round-trip cost + 2 points slippage and re-run. Then hide the last 18 months, re-tune on the rest, and run that hidden slice once. If either step turns green to red, you found the truth *before* the market charged you for it.

# Performance Metrics That Matter

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## Why this matters — the pro vs retail gap this closes

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Ask a retail trader how his strategy performs and he'll quote one number: return, or win rate. Both are near-useless alone. A 90%-win-rate system can blow up (nine Rs 1,000 wins, one Rs 20,000 loss). A 300% backtest return means nothing if it took a 70% drawdown to get there — no human holds through that, and leverage would have killed the account first. Pros read a *panel* of metrics that together answer the only questions that matter: **How much do I make per rupee of risk? How bad does it get? Will I survive the bad stretch? Is the edge real or luck?** This chapter defines the metrics, shows how to compute them from a trade log, and explains why drawdown and expectancy sit at the top of the hierarchy.

## The essentials — the metrics, precisely

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**Return & risk-adjusted return - CAGR** (compound annual growth rate):  $(\text{Ending}/\text{Starting})^{(1/\text{years})} - 1$ . The headline, but blind to risk. - **Sharpe ratio**:  $(\text{annualised return} - \text{risk-free}) / \text{annualised volatility of returns}$ . Return per unit of *total* volatility. Use India's risk-free ~6.5–7% (T-bill / repo neighbourhood, 2026 — *verify*). Sharpe > 1 is good, > 2 is excellent and rare for retail. - **Sortino ratio**: like Sharpe but divides by *downside* deviation only (volatility of negative returns). Fairer, because upside "volatility" isn't a problem. Usually higher than Sharpe. - **Calmar ratio**:  $\text{CAGR} / \text{max drawdown}$ . Return per unit of worst pain. Calmar > 1 is respectable; institutions love it because it speaks to survivability.

**Drawdown - Max drawdown (MDD)**: the largest peak-to-trough fall in equity, in %. This is the metric that decides whether you *survive*. A 50% drawdown needs a 100% gain just to recover. Most retail traders quit (or get margin-called) somewhere around 20–30%.

**Trade-level quality - Win rate**: % of trades that profit. Alone, meaningless — must be read with payoff. - **Avg win / Avg loss (payoff ratio)**: average winning trade ÷ average losing trade. - **Expectancy**: the master number — average rupees (or R) per trade:  $(\text{Win\%} \times \text{Avg Win}) - (\text{Loss\%} \times \text{Avg Loss})$ . If this isn't positive *after 2026 costs*, nothing else matters. - **Profit factor**:  $\text{gross profit} / \text{gross loss}$ . > 1 is profitable; 1.3–1.6 is solid; > 2 is suspiciously good (check for overfitting). - **Exposure**: % of time capital is actually in the market. High return at 5% exposure is very different from the same return fully invested — it affects capital efficiency and how you'd deploy idle cash.

**The hierarchy**. Expectancy and max drawdown outrank everything. A positive net expectancy says the edge exists; a tolerable drawdown says you'll still be trading when it pays off. CAGR without drawdown context is a boast, not information.

## Worked example — reading one strategy's numbers

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You backtest the Bank Nifty opening-range system (after the Chapter-3 honesty fixes) over 4 years on Rs 5,00,000 starting capital. The trade log yields:

- Trades: 480 | Wins: 197 (41.0%) | Losses: 283 (59.0%)
- Avg win (net of 2026 costs): **Rs 9,800** | Avg loss (net): **Rs 6,500**
- Ending equity: Rs 9,20,000 | Worst peak-to-trough equity fall: Rs 5,90,000 → Rs 4,60,000

**Compute the panel:**

- **Payoff ratio** =  $9,800 / 6,500 = 1.51$ .
- **Expectancy** =  $0.41 \times 9,800 - 0.59 \times 6,500 = 4,018 - 3,835 = +Rs\ 183$  per trade. Positive but *thin* — over 480 trades that's ~Rs 87,800 gross of the picture, and it means a small rise in costs or slippage flips it negative. Fragile edge.
- **Profit factor** = gross profit / gross loss =  $(197 \times 9,800) / (283 \times 6,500) = 19,30,600 / 18,39,500 = 1.05$ . Barely above 1 — you're working very hard for a razor-thin edge.
- **CAGR** =  $(9,20,000 / 5,00,000)^{(1/4)} - 1 = (1.84)^{0.25} - 1 \approx **16.5%**$ . Attractive on its own.
- **Max drawdown** =  $(5,90,000 - 4,60,000) / 5,90,000 = 22.0\%$ . Painful — many would quit here.
- **Calmar** =  $16.5\% / 22.0\% = 0.75$ . Below 1: you're taking more drawdown than annual return. Marginal.
- **Sharpe** (assume annualised return 16.5%, annualised vol 18%, risk-free 6.5%) =  $(16.5 - 6.5) / 18 = 0.56$ . Modest — a lot of noise per unit of return.

**Verdict:** the 16.5% CAGR looks great in isolation, but profit factor 1.05, expectancy +Rs 183, Calmar 0.75 and Sharpe 0.56 together say: *thin, fragile edge with drawdowns bigger than its annual return*. One regime shift or a cost hike sinks it. The full panel told the truth the CAGR hid.

## How pros do it / common mistakes

**How pros do it:** - **Read the panel, never one metric.** CAGR + Sharpe/Sortino + max DD + Calmar + expectancy + profit factor, always together. - **Anchor on drawdown and expectancy first.** "Can I survive the worst stretch, and is each trade net-positive after 2026 costs?" - **Read the equity curve visually:** slope (return), smoothness (consistency), and the depth *and length* of underwater periods. A curve that's flat/underwater for 14 months will break your discipline even if it ends higher — length of pain matters as much as depth. - **Compare metrics IS vs OOS.** If Sharpe halves out-of-sample, the in-sample number was fitted. - **Treat profit factor > 2 or a drawdown that looks too small as a red flag**, not a trophy — usually overfitting or a hidden bias.

**Common mistakes / red flags:** - Quoting win rate alone ("I'm right 70% of the time") while ignoring that losses are 4× the wins. - Celebrating CAGR while ignoring a 40% drawdown no human would hold through. - Computing metrics on *gross* returns — in 2026 India, costs are the difference between +Rs 183 and -Rs 500 per trade. - Ignoring the *duration* of drawdowns; a two-year flat patch ends most trading careers. - Too-small a sample (30 trades) making every ratio meaningless noise.

## Checklist / drill

**Metrics review checklist — before trusting any strategy:** - ☐ Expectancy computed *net of 2026 costs* and clearly positive with margin. - ☐ Max drawdown known, in %, and within my personal survivable limit (define it — e.g. 20%). - ☐ Sharpe and Sortino computed with a real India risk-free (~6.5–7%, *verify*). - ☐ Calmar > 1 (or a conscious reason to accept less). - ☐ Profit factor between ~1.3 and ~2 (below = weak, above = check for overfitting). - ☐ Metrics compared IS vs OOS; no metric collapses out-of-sample. - ☐ Equity curve inspected for drawdown *depth and length* and underwater duration.

**Drill:** Export your last 50–100 trades to a sheet. Compute win%, avg win, avg loss, expectancy, profit factor, and max drawdown of the running equity. Then plot the equity curve and mark the deepest and *longest* underwater stretch. Ask honestly: would you have kept trading through that stretch with real money? If the answer is no, your problem isn't the strategy — it's that you were about to trade a drawdown you can't emotionally afford.

# Systematic Strategy Families

## Why this matters

Retail traders collect setups; pros collect *strategy families* and the regimes each one survives in. A candlestick pattern or an RSI cross is a signal — a strategy family is a whole philosophy about *why* an edge exists and *when* it stops working. You already know how to read a chart and price an option. What separates a professional systematic trader is knowing that there are only about half a dozen durable sources of edge, that each one is really a bet on market *regime*, and that running the wrong family in the wrong regime is how accounts die slowly. Indian markets are not the S&P 500: Nifty and Bank Nifty trend hard in some years and chop violently in others, single stocks gap on results, and liquidity is concentrated in a handful of names. Picking the family that fits the instrument and the regime is 80% of the job.

## The essentials

There are six workhorse families. Each is a bet on a different statistical property of returns.

| Family                     | Core bet                                   | Signal example                                 | Works when                   | Dies when                            | India fit                                                   |
|----------------------------|--------------------------------------------|------------------------------------------------|------------------------------|--------------------------------------|-------------------------------------------------------------|
| Trend-following            | Prices auto-correlate over weeks/months    | Price > 200-DMA; 50/200 cross                  | Strong directional regimes   | Range-bound chop                     | Nifty/Bank Nifty index futures, commodity (MCX crude, gold) |
| Momentum (cross-sectional) | Recent winners keep winning vs peers       | Rank Nifty 100 by 6-mo return, long top decile | Broad dispersion             | Sharp reversals/junk rallies         | Nifty 500 stock universe, monthly rebalance                 |
| Mean-reversion             | Short-term over-extension snaps back       | RSI(2) < 10 near support                       | Range/high-liquidity regimes | Trending breakouts, news gaps        | Index intraday, large-cap stocks                            |
| Breakout                   | Range expansion continues                  | Close above 20-day high (Donchian)             | Volatility expansion         | False-break chop                     | Bank Nifty, stock futures on results                        |
| Pairs / stat-arb           | Two related prices revert to a spread      | Z-score of spread > 2                          | Stable relationships         | Structural break (one name re-rates) | HDFCBANK/ICICIBANK, TCS/INFY, PSU banks                     |
| Volatility                 | Vol is mean-reverting & has a risk premium | India VIX percentile, IV vs realised           | Vol clustering               | Vol regime shifts                    | Options on Nifty (covered in the options book)              |

Two India-specific realities shape all of this. First, **cost drag is brutal**: STT (from 01-Apr-2026: equity intraday 0.025% on sell, futures ~0.05% on sell, options ~0.15% on premium sell), plus exchange txn charges, 18% GST on brokerage+txn, stamp duty and SEBI fees. A mean-reversion system that trades 200 times a month can have a real edge on paper and still lose after costs. Trend and cross-sectional momentum trade rarely, so they survive costs far better. Second, **regimes flip**: 2021 and 2023 rewarded trend; 2022 and much of 2018-

19 punished it and rewarded mean-reversion. Pros run *multiple* uncorrelated families so that the drawdown of one is funded by another (next chapter).

A simple cointegration idea for pairs: two stocks A and B are *cointegrated* if some linear combination  $A - \beta \cdot B$  is stationary (mean-reverting) even though A and B each wander. You estimate  $\beta$  by regressing A on B (hedge ratio), form the spread, standardise it to a z-score, and trade the z-score back to zero. You are not betting on direction — you are betting the *relationship* holds. The danger is a structural break: if one bank gets an RBI action or a merger (think HDFC twins in 2023), the spread never reverts and you bleed.

*All rules and rates as of July 2026 — verify on NSE / your broker / SEBI, because they change.*

## Worked example

---

**Mean-reversion, Nifty 50 index (spot 24,000 basis, one lot future = 25 units, tick ₹0.05).** Rule: on the daily chart, buy one Nifty future when RSI(2) closes below 10 *and* price is above the 200-DMA (only fade dips in an uptrend); exit when the close is back above the 5-DMA. Suppose Nifty dips to 23,650, RSI(2) prints 8, 200-DMA is 22,900 — trade is on. Entry 23,650; two days later it closes at 23,880 above the 5-DMA — exit.

- Gross move: 230 points  $\times$  25 = **₹5,750** per lot.
- Costs (round trip, indicative): STT on futures sell 0.05% of  $(23,880 \times 25 = ₹5,97,000) \approx$  **₹298**; brokerage ₹20 + ₹20 = ₹40; exchange txn  $\sim 0.0019\%$  both sides  $\approx$  ₹22; GST 18% on  $(₹40 + ₹22) \approx$  ₹11; stamp  $\sim 0.002\%$  buy  $\approx$  ₹12; SEBI  $\sim$  ₹1. Total  $\approx$  **₹384**.
- Net  $\approx$  **₹5,366** per lot. Margin blocked  $\sim$  ₹1.2–1.4 lakh, so  $\sim 4\%$  on margin in two days — but only because RSI(2) < 10 with a 200-DMA filter is genuinely rare (a handful of trades a month). Strip the trend filter and you catch falling knives.

**Momentum rule (cross-sectional).** Universe = Nifty 100. Monthly: rank stocks by their trailing 6-month return *skipping the last 21 days* (the skip avoids short-term reversal). Go long an equal-weighted basket of the top 10, hold one month, rebalance. This trades  $\sim 12$  times a year per slot, so STT/GST barely dents it. In India this has historically beaten buy-and-hold in dispersion-rich years but suffered "momentum crashes" when beaten-down junk rips (e.g., recovery months). The skip-month and a simple market-regime filter (only run it when Nifty > 200-DMA) tame the worst crashes.

## How pros do it / common mistakes

---

- **Pros match family to instrument and cost.** High-turnover mean-reversion goes on low-cost, high-liquidity index futures — never on illiquid mid-caps where slippage eats the edge.
- **They add a regime filter, not more indicators.** One filter (200-DMA, or VIX percentile) that switches a family on/off beats stacking five oscillators.
- **They diversify across families.** Trend + mean-reversion are naturally negatively correlated; running both smooths the equity curve.
- **Classic retail errors:** running mean-reversion into a trend ("it's oversold" all the way down); backtesting a pairs trade over one stable window and ignoring the break risk; ignoring costs so an intraday scalp that "wins" on the chart loses in the ledger; and curve-fitting RSI thresholds to one year of Nifty data.
- **Red flags:** an edge that only appears with fees set to zero; a pair whose cointegration p-value is borderline; a breakout system with a >60% false-break rate on the chosen instrument.

## Checklist / drill

---

Before deploying any systematic strategy, answer:

1. Which **family** is this, and what statistical property is the bet?
2. In which **regime** does it make money, and what filter turns it off otherwise?
3. Does the edge survive **real Indian costs** (STT dated 01-Apr-2026, GST, exchange, stamp) at my turnover?
4. Is the **instrument liquid** enough that slippage < edge? (Prefer index futures / top F&O names.)
5. For pairs: is the pair **cointegrated out-of-sample**, and what's my break-risk stop?

**Drill:** Take 3 years of daily Nifty data. Code the RSI(2)<10 + above-200-DMA long rule, then the same rule *without* the 200-DMA filter. Compare number of trades, win rate, and net-of-cost P&L. Seeing the filter turn a losing system into a profitable one — purely by respecting regime — is the lesson.



# Position Sizing & Portfolio of Strategies

## Why this matters

You can be right on direction and still go broke. Retail traders obsess over *entries*; professionals know that **position sizing and portfolio construction dominate long-run returns and survival**. Two traders with the identical signal — one sizing 2% risk per trade, the other going "full margin" — have completely different fates: one compounds, the other blows up on the first 6-loss streak that any real edge inevitably produces. In Indian F&O, where SPAN+Exposure margin is fully upfront and peak-margin is enforced, sizing is also a *hard constraint*: you literally cannot deploy what the margin system won't let you. And once you run more than one strategy (previous chapter), the question shifts from "how big is this trade" to "how much of my capital does each *system* get, given how they correlate." That portfolio question is where the free lunch of diversification lives.

## The essentials

**Fixed-fractional (risk-per-trade).** Risk a fixed fraction  $f$  of equity per trade. Position size = (equity  $\times$   $f$ )  $\div$  (per-unit risk to stop). Typical  $f$  = 0.5%–2%. This auto-scales: you bet more as you win, less as you lose. It is the retail-safe default.

**Volatility targeting.** Instead of fixed rupees, size so each position contributes the same *risk*. Target an annualised portfolio vol (say 15%). Position notional = (target vol  $\div$  instrument's realised vol)  $\times$  capital  $\div$   $n$ . When Bank Nifty's vol doubles, you halve the lots — this keeps risk stable across calm and stormy regimes, and is how most pro futures books run.

**Kelly criterion.** The growth-optimal fraction. For a simple bet,  $f^* = \text{edge}/\text{odds} = p - (1-p)/b$  where  $p$  = win prob,  $b$  = win/loss payoff ratio. Kelly *maximises long-run compounding* but is wildly volatile — full Kelly routinely draws down 50%+. Because your estimated edge is uncertain and Kelly is brutal when you overestimate it, pros use **fractional Kelly**: half-Kelly or quarter-Kelly. Half-Kelly keeps ~75% of the growth for ~half the volatility. Nobody serious trades full Kelly.

**Correlation between strategies.** Diversification benefit depends on correlation  $\rho$ . Two strategies each with vol  $\sigma$ , combined 50/50, have portfolio vol  $\sigma \cdot \sqrt{(1+\rho)/2}$ . At  $\rho=0$  that's  $\sigma \times 0.71$  (29% vol reduction for free). At  $\rho=1$  you get nothing. This is *the* reason to run trend + mean-reversion together — they're often negatively correlated.

**Capital allocation & risk parity.** Naive allocation splits *capital* equally; **risk parity** splits *risk* equally, so each strategy contributes the same volatility to the book. A low-vol pairs system then gets more capital than a high-vol momentum system, so one doesn't dominate the P&L.

| Method           | Sizes on            | Best for                               | Watch out                              |
|------------------|---------------------|----------------------------------------|----------------------------------------|
| Fixed-fractional | Distance to stop    | Single discretionary/systematic trader | Ignores changing vol                   |
| Vol targeting    | Realised volatility | Futures/multi-instrument books         | Needs a vol estimate; lags jumps       |
| Fractional Kelly | Edge & odds         | Known, stable edge                     | Overestimating edge $\rightarrow$ ruin |
| Risk parity      | Each strategy's vol | Portfolio of systems                   | Correlations drift/spike in crises     |

## Worked example

**Capital ₹15,00,000. Fixed-fractional at  $f = 1\%$  → ₹15,000 risk per trade.** Bank Nifty future, lot = 15 units, spot 52,000. Your system's stop is 300 points away → per-lot risk =  $300 \times 15 = ₹4,500$ . Lots =  $₹15,000 \div ₹4,500 = 3.3 \rightarrow 3$  lots. SPAN+Exposure margin  $\approx ₹1.6\text{--}1.9$  lakh/lot, so 3 lots  $\approx ₹5.4$  lakh margin — comfortably within ₹15 lakh. Note the *risk* (₹13,500) is far below the *margin* (₹5.4 lakh): margin is a collateral requirement, risk is what you actually stand to lose to your stop. Confusing the two is a classic error.

**Fractional Kelly check.** Suppose the system wins 45% of the time with a 2:1 payoff ( $b=2$ ). Full Kelly  $f = 0.45 - 0.55/2 = 0.45 - 0.275 = 0.175$  (17.5%). That is *insanely aggressive* — a 17.5% risk-per-bet system will have gut-wrenching 50%+ drawdowns. Quarter-Kelly =  $\sim 4.4\%$ , still above our 1% choice. So 1% is conservative relative to Kelly — deliberately, because the 45%/2:1 estimate is itself uncertain.

**Portfolio of two systems, risk parity.** System A (trend, index futures) vol 20% p.a.; System B (mean-reversion) vol 12% p.a.; correlation  $\rho = -0.2$ . Risk-parity weights  $\propto 1/\text{vol} \rightarrow A : B = (1/20):(1/12) = 0.375 : 0.625$ . Combined vol with  $\rho = -0.2$  is roughly  $\sqrt{(w_A^2 \cdot 20^2 + w_B^2 \cdot 12^2 + 2 \cdot w_A \cdot w_B \cdot 20 \cdot 12 \cdot (-0.2))} \approx \sqrt{(56.25 + 56.25 - 22.5)} \approx \sqrt{90} \approx 9.5\%$  p.a. — lower than either standalone system. Same expected return, one-third less volatility: that is the diversification free lunch, and why allocating across uncorrelated families beats leveraging one.

## How pros do it / common mistakes

- **Pros size on risk, not on margin or "conviction."** They compute rupees-at-risk to the stop first, then check margin as a constraint.
- **They cap total portfolio risk**, e.g. no more than 6% of equity at risk across all open positions, and a per-strategy cap so no single system can sink the book.
- **They use fractional Kelly** and treat their estimated edge as optimistic. If in doubt, they *underbet*: half-Kelly loses little growth, full-Kelly plus a bad estimate is ruin.
- **They monitor correlation, knowing it spikes in crises.** Diversification that shows  $\rho=0$  in calm markets can jump toward 1 in a crash — so they stress-test with correlations forced to +0.5.
- **Retail mistakes:** sizing to max margin ("3 lots because I can afford 3 lots"); adding to losers (anti-Kelly); running two "different" strategies that are secretly the same momentum bet ( $\rho \approx 0.9$ , zero diversification); ignoring that a 2% edge with 5% sizing blows up on a normal losing streak.
- **Red flags:** any single trade risking  $>2\text{--}3\%$  of equity; total open risk you can't state in rupees; a "diversified" book where all systems drop together.

## Checklist / drill

1. What is my **rupees-at-risk per trade** (equity  $\times f \div$  per-unit stop distance)?
2. Is that within my **per-trade cap** ( $\leq 2\%$ ) and **total portfolio cap** ( $\leq 6\%$ )?
3. Does **margin** allow the position — and am I not confusing margin with risk?
4. If sizing by Kelly, am I using  **$\leq$  half-Kelly** and treating my edge estimate as optimistic?
5. Across strategies, what are the **pairwise correlations**, and does the book survive them all going to +0.5?

**Drill:** Take two of your backtested systems' daily-return series. Compute each one's volatility and their correlation. Build (a) equal-capital and (b) risk-parity portfolios; compare portfolio vol and max drawdown. Then re-run with correlation forced to +0.5 to see how much of the "diversification" was real. Feeling the equity curve smooth out at low correlation — and roughen at high — is the entire lesson.

# Python for Algo Trading

## Why this matters

The gap between a retail "backtest" in TradingView's strategy tester and a professional research workflow is enormous, and it is mostly about *reproducibility and honesty*. TradingView hides your assumptions; a Python notebook forces you to state every one — data source, costs, slippage, lookahead — in code you can re-run and audit. Pros don't trust a strategy they can't reproduce from raw OHLC to equity curve on their own machine. Python is the lingua franca because pandas makes OHLC manipulation trivial, the ecosystem (yfinance, nsepy, broker SDKs, backtesting.py, vectorbt) is free, and vectorised code lets you test 20 years of Nifty in milliseconds. This chapter is about building a *minimal, reproducible* research loop: get data, compute a signal, backtest it with realistic Indian costs, and plot the equity curve — the skeleton every serious system starts from. (Live execution and SEBI's algo rules are the next chapter; this one stays in research.)

## The essentials

### Data sources (India).

| Source                | Gets you                                                                                                  | Notes                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| yfinance              | Free daily OHLC; <code>^NSEI</code> (Nifty), <code>^NSEBANK</code> (Bank Nifty), <code>RELIANCE.NS</code> | Great for research; adjust for splits; not tick data          |
| nsepy / jugaad-data   | NSE historical, indices, F&O                                                                              | Community-maintained; NSE changes break them — pin versions   |
| Broker historical API | Zerodha Kite, Upstox intraday/minute candles                                                              | Auth required; rate-limited; the real source for live systems |

**The stack.** `pandas` for the OHLC DataFrame (a DatetimeIndex plus Open/High/Low/Close/Volume). Compute indicators as vectorised column operations ( `.rolling()` , `.ewm()` , `.pct_change()` ) — no Python loops. Backtest with either `backtesting.py` (event-driven, readable, easy costs) or `vectorbt` (fully vectorised, blazing fast for parameter sweeps). Plot the equity curve with `matplotlib` .

**Costs must be modelled.** A backtest without Indian costs is a lie. Bake in STT (from 01-Apr-2026: equity intraday 0.025% on sell, futures ~0.05% on sell), brokerage (~₹20/order), exchange txn, 18% GST on brokerage+txn, stamp duty, plus **slippage** (assume you don't get the close). In `backtesting.py` , `commission=` covers a percentage; add a slippage buffer yourself.

**The cardinal sin: lookahead bias.** Compute a signal on bar  $t$ , act on bar  $t+1$ 's open. If you compute an indicator using today's close and also "buy at today's close," you've peeked. Shift signals by one bar.

*Library behaviour and NSE data endpoints as of July 2026 — pin package versions and re-verify; free data sources break often.*

## Worked example

A minimal, reproducible SMA-crossover backtest on Nifty daily, with costs. Run in a fresh venv: `pip install yfinance backtesting pandas matplotlib` .

```

# nifty_sma_backtest.py - reproducible research skeleton (July 2026)
import yfinance as yf
import pandas as pd
from backtesting import Backtest, Strategy
from backtesting.lib import crossover

# 1. DATA: Nifty 50 daily OHLC
data = yf.download("^NSEI", start="2015-01-01", end="2026-06-30",
                    auto_adjust=True, progress=False)
data = data[["Open", "High", "Low", "Close", "Volume"]].dropna()

# 2. INDICATOR helper (vectorised)
def SMA(series, n):
    return pd.Series(series).rolling(n).mean()

# 3. STRATEGY: long-only 50/200 SMA cross (trend-following)
class SmaCross(Strategy):
    n1, n2 = 50, 200
    def init(self):
        self.sma1 = self.I(SMA, self.data.Close, self.n1)
        self.sma2 = self.I(SMA, self.data.Close, self.n2)
    def next(self):
        # acts on NEXT bar's open → no lookahead
        if crossover(self.sma1, self.sma2):
            self.buy()
        elif crossover(self.sma2, self.sma1):
            self.position.close()

# 4. BACKTEST with realistic cost drag (commission ~ round-trip proxy incl. STT/GST/slippage)
bt = Backtest(data, SmaCross, cash=1_500_000,
               commission=0.0006, trade_on_close=False)
stats = bt.run()
print(stats[["Return [%]", "Sharpe Ratio", "Max. Drawdown [%]",
             "# Trades", "Win Rate [%]"]])

# 5. EQUITY CURVE
bt.plot(filename="nifty_sma_equity.html", open_browser=False)

```

What this teaches with real numbers: over ~11 years, a 50/200 cross on Nifty produces very *few* trades (roughly 10–20), so the 0.06% cost proxy barely dents it — a trend system's low turnover is its cost advantage (contrast a daily mean-reversion system trading 200×/year, where the same 0.06% compounds into a big drag). The printed Sharpe and Max Drawdown are your honest scorecard. Change `n1`, `n2` and re-run to feel how *sensitive* results are — if 50/200 works but 48/210 collapses, you've curve-fit. To sweep parameters fast, port the same signal to `vectorbt` and vectorise across a grid of (n1, n2) in one call.

For an intraday variant you'd pull minute candles from the Kite historical API instead of yfinance, and raise the cost/slippage assumption sharply (intraday STT + more fills).

## How pros do it / common mistakes

---

- **Pros separate research from execution.** The backtest above never touches a live order. Signal logic is reused; execution is a separate, tested layer (next chapter).
- **They pin versions and cache data** so a result is reproducible months later — `pip freeze > requirements.txt`, save the raw CSV.
- **They model costs and slippage pessimistically.** If the edge survives double the expected costs, it might be real.
- **They test out-of-sample.** Fit parameters on 2015–2021, validate untouched on 2022–2026. In-sample-only results are fiction.
- **Retail mistakes:** lookahead bias (buying at the close using the close); zero-cost backtests; over-fitting to one parameter set; ignoring survivorship bias (a stock universe that excludes delisted names flatters returns); trusting a stellar equity curve built on 4 trades.
- **Red flags:** Sharpe > 3 on daily data; an equity curve with no drawdowns; results that vanish when you shift signals by one bar (that was lookahead).

## Checklist / drill

---

1. Is my data source **pinned and cached**, so this is reproducible?
2. Do signals **act on  $t+1$** , with no lookahead?
3. Are **STT (01-Apr-2026), GST, exchange, stamp AND slippage** all modelled?
4. Did I validate **out-of-sample**, not just in-sample?
5. Is the result **robust** to small parameter changes (not curve-fit)?

**Drill:** Run the script above. Then (a) set `commission=0` and note how much the reported return inflates — that gap is your cost reality; (b) change 50/200 to 48/210 and see if the edge holds; (c) deliberately introduce lookahead ( `trade_on_close=True` with a close-based signal) and watch the Sharpe jump — that jump is exactly the bias pros hunt for and kill.

# Execution, Broker APIs & the SEBI Retail-Algorithm Framework (2026)

## Why this matters

A backtest with a great Sharpe and a live account that loses money are separated by one thing amateurs ignore: **execution**. Slippage, latency, partial fills and rejected orders quietly erode edge, and a system that trades 100 times a day can lose its entire theoretical profit to bad fills. Worse, since **01-Apr-2026 SEBI's retail algo framework is mandatory** — and getting it wrong is no longer a "your P&L" problem, it's a *compliance* problem that can freeze your account or expose you to penalties. Pros treat execution and compliance as first-class engineering, not an afterthought. This chapter covers the execution layer (routing, paper trading, slippage/latency), the Zerodha Kite Connect flow as the concrete Indian example, and — most importantly — the 2026 SEBI rules you must obey before a single automated order goes out. Compliance first; cleverness second.

## The essentials

**Order routing & types.** Your order travels app/API → broker OMS/risk checks → exchange (NSE/BSE) matching engine → confirmation back. Use the right type: **MARKET** (guaranteed fill, unknown price — dangerous in illiquid/fast markets), **LIMIT** (known price, maybe no fill), **SL / SL-M** (stop-loss trigger). For anything beyond the most liquid names, prefer LIMIT to control slippage.

**Paper trade first.** Before real capital, run the *live signal* against live data placing simulated orders (or use the broker's sandbox). This surfaces the gap between backtest fills and reality — the single most sobering step for a new algo trader.

**Slippage & latency.** Slippage = (fill price – intended price). Latency = time from signal to fill (network + broker + exchange). For retail on a broker REST API, expect tens to hundreds of milliseconds — you are *not* competing with co-located HFT, so **do not design strategies whose edge lives inside a few milliseconds**. Choose signals that tolerate realistic fills.

**Zerodha Kite Connect flow (concrete India example).**

| Step            | What happens                                                                                                                                                                     |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Auth            | App <code>api_key</code> / <code>api_secret</code> ; user login → <code>request_token</code> → exchange for a daily <code>access_token</code> (expires each day — re-auth daily) |
| Place           | <code>kite.place_order(variety, exchange, tradingsymbol, transaction_type, quantity, order_type, product, price)</code> → returns <code>order_id</code>                          |
| Modify / Cancel | <code>kite.modify_order(...)</code> , <code>kite.cancel_order(...)</code> by <code>order_id</code>                                                                               |
| Postbacks       | Webhook pushes order status updates (COMPLETE/REJECTED/etc.) — event-driven, don't poll                                                                                          |

```

from kiteconnect import KiteConnect
kite = KiteConnect(api_key="xxx")
kite.set_access_token(access_token)          # from daily login flow
oid = kite.place_order(variety=kite.VARIETY_REGULAR,
                        exchange=kite.EXCHANGE_NFO, tradingsymbol="BANKNIFTY26JUL52000CE",
                        transaction_type=kite.TRANSACTION_TYPE_BUY, quantity=15,
                        order_type=kite.ORDER_TYPE_LIMIT, product=kite.PRODUCT_MIS, price=180.5)

```

**The mandatory SEBI Retail Algo Framework (effective 01-Apr-2026).** Every algo order must carry an **exchange-registered Algo-ID** so the exchange can tag and trace it. **Open/unauthenticated APIs are BANNED** — retail algos may run *only* through a registered broker's authenticated API. **Third-party algo vendors must tie up with a registered broker** and register their algo with the exchange. **Orders above 10 per second** from a retail participant require specific exchange approval/registration (treated like higher-frequency activity). **The broker is responsible** for algos operating on its platform — meaning they enforce these rules on you. Static/DIY logic below the order-rate threshold, through your own broker's authenticated API, is the retail-friendly path; anything faster or offered to others crosses into registration territory.

*All rules as of July 2026 — the SEBI algo framework, STT and margin rules change; verify on the SEBI circulars, NSE/BSE notices and your broker's algo policy before deploying.*

## Worked example

You've validated the Bank Nifty trend system from the Python chapter and want it live via Kite Connect.

- **Instrument/costs:** Bank Nifty options, lot = 15. Say the system fires ~8 signals/day. That's ~16 orders/day round-trip — **well under 10 orders/second**, so no special exchange registration for order-rate; but you still need a **valid Algo-ID** tagged to every order, obtained via your broker's algo-registration process (Zerodha's static-algo registration), because from 01-Apr-2026 automated orders without one can be rejected/flagged.
- **Compliance path:** register the strategy with the broker → receive Algo-ID → route orders through the authenticated Kite Connect API (never a scraped/open API) → keep audit logs.
- **Slippage reality check:** backtest assumed fills at ₹180.50. Live, over 200 trades you observe an average fill of ₹181.20 on entries and ₹179.60 on exits — ~₹1.10 total slippage per unit × 15 × 200 = **₹3,300** of edge quietly gone, *before* STT (options ~0.15% on premium sell from 01-Apr-2026), brokerage, exchange txn, 18% GST and stamp. If the backtested edge was ₹4,000, you're barely breakeven — which is *exactly* why you paper-trade to measure real slippage before committing capital.
- **Latency:** signal→fill ~150 ms on a home connection is fine for a daily/positional trend system; it would be fatal for a "scalp the first tick" idea, so you don't build that idea.

## How pros do it / common mistakes

- **Pros are compliance-first.** They obtain the Algo-ID, use only the broker's authenticated API, stay under the order-rate threshold unless properly registered, and keep immutable logs. They treat SEBI rules as hard constraints, not suggestions.
- **They paper-trade to measure slippage** and only fund a strategy once live fills confirm the backtest edge survives.



- **They build kill-switches and reconciliation** — a max-loss auto-halt, and end-of-day checks that broker positions match the system's expected positions.
- **They handle failures** — rejected orders, token expiry, network drops — with retries and alerts, never assuming a placed order filled.
- **Retail mistakes:** using banned open/unofficial APIs (post-01-Apr-2026 this risks account action, not just rejects); hardcoding a `MARKET` order into an illiquid option and eating huge slippage; ignoring daily `access_token` expiry so the bot silently stops; assuming placement = fill; deploying a millisecond-edge idea on a REST API that will never be that fast; no kill-switch on a runaway loop.
- **Red flags:** any dependency on an unregistered third-party algo without a broker tie-up; sustained order rates near/over 10/sec without exchange registration; live P&L diverging from backtest by more than modelled costs (usually slippage or a bug).

## Checklist / drill

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Pre-deployment compliance & execution gate (as of July 2026 — re-verify current SEBI/exchange rules):

1. Does every automated order carry a valid **exchange Algo-ID** via my broker's registration?
2. Am I routing **only through the broker's authenticated API** (no open/unofficial APIs)?
3. Is my **order rate < 10/sec** — or am I properly exchange-registered for more?
4. Have I **paper-traded** long enough to measure real **slippage & latency**, and does the edge survive them plus full costs?
5. Do I have a **kill-switch, daily token-refresh handling, and position reconciliation**?

**Drill:** Take your validated strategy and run it in **paper mode against live market data for two weeks** without real capital. Log intended price vs simulated fill for every order, compute average slippage per trade, and subtract it (plus STT/GST/exchange/stamp) from the backtest edge. If the net edge disappears, the strategy was never real — you just saved yourself the tuition. Only strategies that survive this go live, and only after the Algo-ID and API-compliance boxes above are all ticked.

# Deploying & Running a Live Algo

## Why this matters

Your backtest showed a beautiful equity curve. That is the easy 20%. The other 80% — the part that separates a hobbyist from a professional — is running the thing live, unattended, without it blowing up your account while you sleep or step away for chai. A backtest is a spreadsheet that never disconnects, never gets a partial fill, never sees SPAN margin spike at 14:55, and never has its VPS reboot for a Windows update mid-position. Live is where all of that happens on the same day.

The retail gap here is brutal. Most retail "algos" are a Python loop on a home laptop with `while True: place_order()`, no logging, no reconciliation, no kill-switch — and a broker API key. When the Wi-Fi drops mid-trade, the trader has an open naked Bank Nifty position and no idea. Pros treat deployment as an engineering problem: assume every component *will* fail, and design so that failure is safe (flat or hedged), not catastrophic. This chapter is that engineering discipline, India-specific.

**A compliance note you cannot skip (as of 2026 — verify on NSE/SEBI, rules change):** Under SEBI's retail algo framework, **mandatory from 01-Apr-2026**, every algo order must carry an **exchange Algo-ID**, **open/unregistered APIs are banned**, and retail algos may only run **through your registered broker's authenticated API**. Orders above **10/second** need exchange approval/registration, and the **broker is responsible** for algos on their platform. Third-party algo vendors must tie up with a registered broker. Translation: you deploy against Zerodha Kite Connect, Dhan, Fyers, etc. with proper auth — not a scraped session token.

## The essentials

Think of a live deployment as five layers, each of which must be independently reliable.

| Layer                 | What it does                               | India-specific detail                                                                                                            |
|-----------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Infra</b>          | Where code runs 24/5                       | VPS/cloud in <b>Mumbai region</b> (AWS ap-south-1 / low-latency co-lo). ~5–20 ms to NSE beats a home line's 60–150 ms + outages. |
| <b>Scheduler</b>      | Starts/stops the strategy on the clock     | Login token refresh <b>before 09:15</b> , square-off timer at <b>15:20</b> , hard stop by <b>15:30</b> .                         |
| <b>Logging</b>        | Immutable record of every decision & order | Every signal, order request, order response, fill, error — timestamped to the millisecond.                                       |
| <b>Monitoring</b>     | Tells you when something breaks            | Heartbeat + alerts to Telegram/phone, not just an email you'll read tomorrow.                                                    |
| <b>Reconciliation</b> | "Does the broker agree with me?"           | Compare <i>my</i> expected positions/orders vs broker's actual, every cycle.                                                     |

**Scheduling.** Indian equity/F&O trades **09:15–15:30 IST**. Your daily flow: (1) a cron/scheduler job at ~08:45 refreshes the broker access token (Kite tokens expire daily — you must re-auth each morning), (2) strategy starts consuming ticks at 09:15, (3) a forced square-off at 15:20 for intraday, (4) everything hard-flat and process idle by 15:30. Never let an intraday algo carry a position past auto-square-off — the broker's RMS will close it at a worse price and charge you for it.

**Logging.** Two streams: a human-readable app log and a structured **orders ledger** (CSV/SQLite/DB row per order with `client_order_id`, timestamp, symbol, side, qty, price, status, broker\_order\_id). The `client_order_id` you generate is your anchor for reconciliation and for tax/audit later.

**Monitoring & kill-switch.** A **heartbeat**: the process writes "alive + P&L + open positions" every 30–60 s to a channel you watch. Two automated cutoffs must exist: - **Max-loss cutoff**: if realized+unrealized P&L breaches a rupee floor, square off everything and **stop trading for the day**. - **Kill-switch**: a manual big red button (a Telegram command or a sentinel file) that flattens and halts, for when *you* see something wrong that the code doesn't.

**Disconnects & partial fills.** Assume the WebSocket tick feed will drop. On disconnect: stop generating new entries, reconnect with backoff, and on reconnect **reconcile before acting**. For orders, never assume "placed = filled." Poll order status / consume postbacks; handle `qty=100 requested, 50 filled` explicitly (you own 1 lot, not 2).

## Worked example

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Strategy: intraday **Bank Nifty** long-straddle-scalp exit logic — but the point here is the *deployment*, so take a simple one: buy 1 lot Bank Nifty futures on a signal, target/stop intraday, forced square-off 15:20. **Bank Nifty futures lot = 15** (verify current lot on NSE — revised periodically). At an index level of 52,000, 1 lot notional  $\approx ₹7,80,000$ ; SPAN+Exposure margin  $\approx ₹1.4\text{--}1.6$  lakh (varies daily — check broker margin calc).

Deployment settings: - **VPS**: AWS ap-south-1, ₹1,200–2,500/month t3.small. Measured latency to broker: 8 ms. - **Max-loss cutoff**: ₹6,000/day (roughly  $0.8 \text{ Bank Nifty point-move} \times 15 \times \text{a few ticks buffer}$ , sized to  $\sim 4\%$  of the margin block). - **Per-trade stop**: 120 points =  $120 \times 15 = ₹1,800$ .

Now the live-vs-backtest reality on one round trip (approx, **STT from 01-Apr-2026** — futures  $\sim 0.05\%$  on **sell** side; verify with broker contract note): - Buy 52,000, sell 52,120 (a +120 winner). Gross =  $120 \times 15 = ₹1,800$ . - Costs: brokerage ₹20 (₹40 both legs, capped) + STT on sell  $\approx 0.05\% \times (52,120 \times 15) \approx ₹391$  + exchange txn + SEBI + stamp + **18% GST** on (brokerage+txn)  $\approx$  another ₹30–60. Realistic total costs  $\approx ₹450\text{--}480$ . - **Net  $\approx ₹1,320$  on a ₹1,800 gross winner**. Your backtest that ignored STT/GST was overstating each winner by  $\sim ₹450$  and understating each loser by the same. Over 200 trades/month that is  $\sim ₹90,000$  of pure friction the backtest hid.

The disconnect scenario that pays for all this plumbing: at 11:32 the tick feed drops for 90 seconds while you're long 1 lot. A naive loop keeps firing on stale prices. The engineered version: heartbeat misses → alert fires → new-entry logic freezes → on reconnect, reconciliation reads broker positions, confirms 1 lot long still open, resumes stop management. No duplicate order, no phantom position.

## How pros do it / common mistakes

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**How pros do it - Paper → small-live → scale.** Run the exact live code against a paper/simulated broker for 2–4 weeks, then live with **1 lot and a tiny max-loss** for 3–4 weeks, only then scale size. The bugs that matter (token refresh, partial fills, reconnect) only appear live. - **Idempotent orders.** Generate a unique `client_order_id`; never re-fire an order just because you didn't see a response — check first. - **Reconcile every loop**, and again at startup (you may restart mid-day into an existing position). - **Fail flat.** Any unhandled exception → square off (or confirm hedged) and halt, rather than continue in an unknown state. - **Separate config from code.** Max-loss, lot size, symbol in a config file you can change without redeploying.

**Common retail mistakes / red flags** - Home laptop + home Wi-Fi as "production." One power cut = open naked position. - No max-loss cutoff — the algo "revenge trades" a bad day to -₹80,000. - Assuming fills; double-sizing on partials. - Backtest with **zero costs** — then live bleeds on STT/GST/slippage. - Hard-coding a broker token that expires at midnight; algo silently dead next morning. - Running an **open/third-party API** algo — now outright non-compliant under the 2026 SEBI framework (verify on SEBI/NSE). - No reconciliation → your code thinks flat, broker RMS auto-squared you at 15:20 at a bad price.

## Checklist / drill

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**Pre-deploy checklist (run before going live with real money):** 1. Broker API registered, Algo-ID/authenticated API in place (SEBI 2026 compliant — verify). 2. VPS in ap-south-1; latency measured; auto-restart on reboot configured. 3. Daily token-refresh job scheduled and tested to succeed before 09:15. 4. Orders ledger writing every request/response with `client_order_id`. 5. Heartbeat posting P&L + positions to a channel on your phone. 6. **Max-loss cutoff** wired and unit-tested to square off + halt. 7. **Kill-switch** (manual flatten) tested end-to-end today. 8. Disconnect handling: reconnect-with-backoff + reconcile-before-acting. 9. Partial-fill logic explicitly handled and tested. 10. Forced square-off at 15:20; process idle by 15:30. 11. Costs (STT/GST/txn/stamp) modeled in the *live* P&L, not just backtest. 12. Startup reconciliation: on restart, read broker positions before trading.

**Drill:** In paper mode, kill your internet for 2 minutes mid-position. Confirm: no duplicate orders fired, the position is correctly reconciled on reconnect, an alert reached your phone, and the max-loss cutoff still armed. If any of those four fail, you are not ready for real money.

*Rules and figures date-stamped to 2026 — always re-verify STT, lot sizes, margins, and the SEBI algo framework on NSE/your broker/SEBI before deploying; they change.*